MIL-N-26206B(USAF)
16 June 1965
SUPERSEDING
MIL-N-26206A(USAF)
18 July 1958

MILITARY SPECIFICATION

NOZZLE, FIRE HOSE, FOAM-WATER, AIRCRAFT FIRE FIGHTING, 60 GPM AND 100 GPM

1. SCOPE

1.1 Scope. This specification covers conventional, aspirating-type, 60- and 100-gpm combination foam and water handline nozzles.

1.2 Classification. Nozzles shall be of the following types, as specified (see 6.2):

Type I - 60 gpm

Type II - 100 gpm.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-H-775	Hose, Rubber or Fabric (Including Tubing), and Fittings, Nozzles, and Strainers, Packaging of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association Standard

NFPA No. 412	Test Procedures for Aircraft Rescue and Fire Fighting Vehicles

FSC 4210

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(Copies of the NFPA standard may be obtained from the National Fire Protection Association, 60 Batterymarch Street, Boston 10, Massachusetts.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Dissimilar metals. The use of dissimilar metals in contact which tend toward electrolytic corrosion (particularly brass, copper, or steel in contact with aluminum or aluminum alloy) shall not be acceptable. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces will be permitted. A suitable insulating material shall be used to separate dissimilar metals. Dissimilar metals are defined in MS33586.

3.3.2 Protective treatment. Materials used in the nozzle construction shall resist, or shall be treated to resist, corrosive and deteriorating effects of foam liquids (see 6.3.1), and other environmental conditions likely to occur during service usage. Any protective coating that will crack, chip, or scale with age or extremes in environmental conditions shall not be used.

3.4 Design and construction. The nozzle shall conform substantially to figure 1 which illustrates a typical design of the nozzles covered by this specification. The nozzle shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Reliability. The nozzle shall have a minimum MCBF (mean-cycle-between-failure) of 16 cycles with a minimum reliability of 0.87 for a 1-cycle mission at a confidence factor of 0.90.

3.4.2 Maintainability. The nozzle shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with the reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection, and for recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determine that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimum number of tools required for maintenance by design practices such

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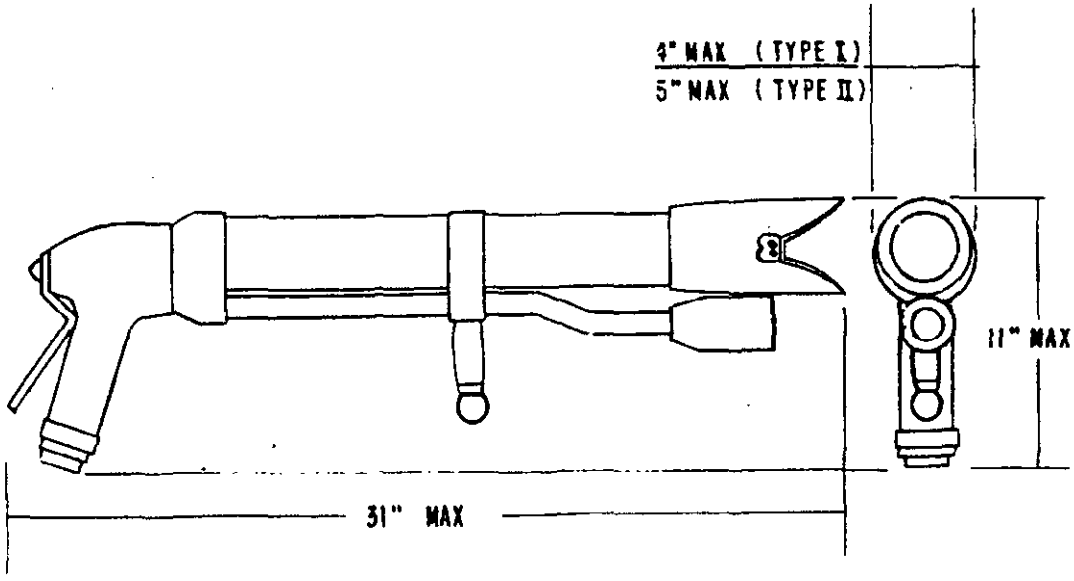


FIGURE 1. Dimensions

as reducing the variety of bolt head sizes to the practical minimum.

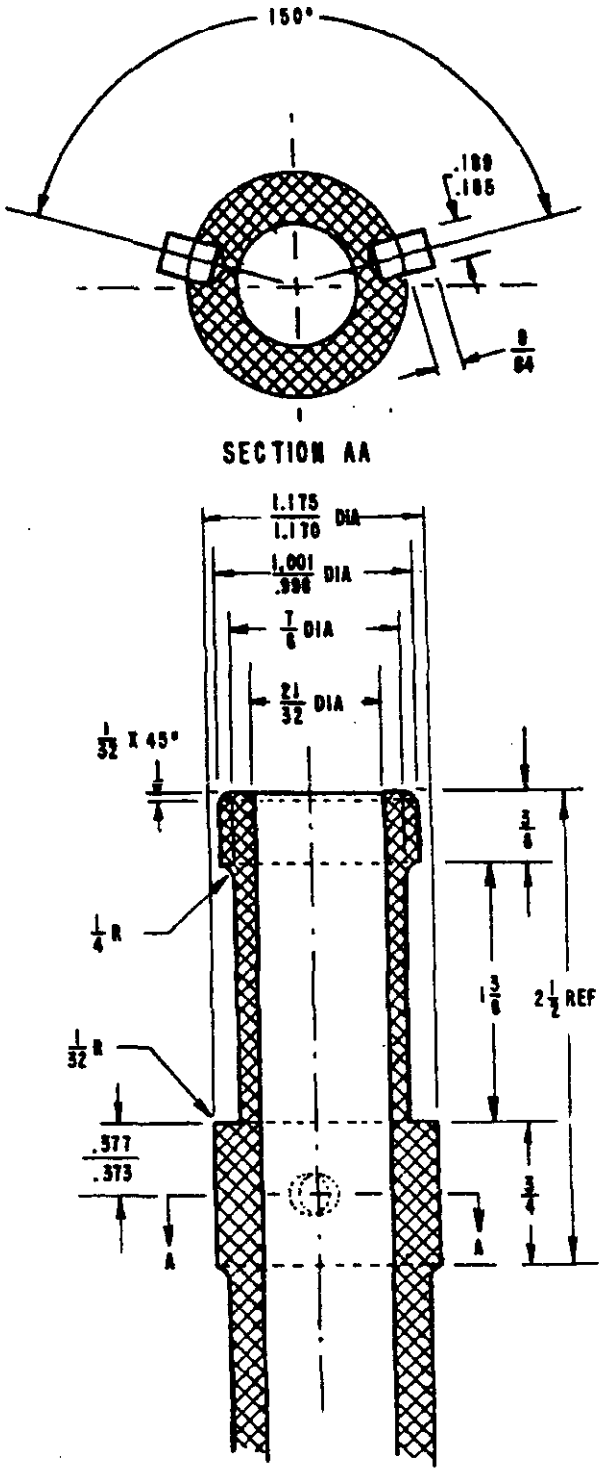
3.4.3 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.4.4 Ease of handling and operation. The nozzle shall be designed and constructed to provide for ease of handling and operation under the conditions specified herein. The nozzle shall be so designed that clothing worn by operating personnel will not obstruct its operation. All controls shall be easily operated by personnel wearing heavy winter gloves or mittens.

3.4.5 Nozzle assembly. All components shall be provided in one assembly, including discharge tips and mixing tubes or chambers, to permit change from water to foam operation or vice versa. The nozzle may be designed to include the following subassemblies.

3.4.5.1 Valve assembly. The valve assembly shall incorporate a recess having a depth of 2.515 ±0.005 inch into which a bayonet applicator joint in accordance with figure 2 can be inserted and secured. The nozzle face shall have a minimum 1-5/8-inch-diameter area free of obstructions around the hole into which the bayonet applicator is inserted. The back end of the nozzle shall be of a pistol-grip design and shall include a nozzle-to-hose swivel connection at the bottom of the handle that will allow unlimited rotation of the nozzle in either direction when attached to the hose. The swivel connection shall incorporate a 1-inch female National Fire Hose thread on the type I nozzle and a 1-1/4-inch female NPSH thread on the type II nozzle. The valve assembly shall include adjustable or self-aligning features to assure ease of operation and leakproof closing. The force required to open and close the valve under operating pressures up to 600 psi shall not exceed 16 pounds. The

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Dimensions in Inches. Unless Otherwise Specified, Tolerences: Fractions ±0.005 Angles ±1/2°

FIGURE 2. Bayonet Nozzle Joint

discharge valve control shall be provided with features to prevent accidental turning on and off when subjected to line pressures and while the nozzle is being handled by fire-fighting personnel. If detents are used to secure the valve operating control in position, bearing surfaces shall be hardened to prevent grooving. The nozzle housing or operating handle shall be permanently stamped or embossed in large letters to indicate the various positions for obtaining the different discharge patterns.

3.4.5.1.1 Strainer. A strainer, if required, shall be provided to protect all orifices subject to clogging by foreign particles 1/8-inch or less in diameter. The strainer shall be installed within the confines of the nozzle assembly and shall be of adequate size to provide unrestricted flow through the nozzle during extended periods of fire-fighting operations. The strainer shall include features to facilitate cleaning and replacement.

3.4.5.2 Extension assemblies and tips. The nozzle or assemblies shall incorporate features which will permit the agents to be discharged as straight-stream water, fan-shaped water stream, water-fog, solid-stream foam, or variable range dispersed foam (see 6.3.2 through 6.3.6). When the nozzle is completely assembled, all components for the various discharges shall form a complete unit; however, removal of components for utilization of the bayonet applicator shall be possible.

3.5 Performance

3.5.1 Pressure and discharge capacities. The maximum operating pressure at the nozzle shall not exceed 110 psig. Features shall be so provided that operating pressures will stay within the maximum allowed when changing from water to foam discharge, and vice versa, with the pump pressure remaining constant.

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The nozzle shall exhibit the following minimum discharge capacities:

	Type I (GPM)	Type II (GPM)
Straight-stream water	50	50
Water-fog	50	50
Fan-shaped water stream	50	50
Foam (water rate)	60	100

3.5.2 Discharge range. The maximum range of discharge (see 6.3.7) at design operating pressure shall be not less than the following:

	Type I (Feet)	Type II (Feet)
Straight-stream water	70	70
Water-fog	12	12
Fan-shaped water stream	25	25
Solid-stream foam	45	60
Dispersed foam, variable (Maximum dispersed to straight stream)	20 thru 45	20 thru 60

3.5.3 Water rate. The average water rate (see 6.3.8) shall exceed 0.20 gpm per square foot of ground pattern area (see 6.3.11) for all discharge conditions.

3.5.4 Drainage time. The drainage time (see 6.3.9) of the foam shall be at least 6 minutes.

3.5.5 Expansion. The expansion (see 6.3.10) of the foam shall be at least 9.

3.5.6 Variable range. The nozzle shall discharge foam for effective extinguishment of pooled gasoline fires from a maximum distance (see 3.5.2) to a minimum distance of 6 feet without excessive agitation of burning fuel. Excessive agitation occurs when, upon impact by the foam agent, the burning fuel is scattered. Dispersed foam discharge shall be in a flattened pattern. While discharging foam at a maximum dispersion setting, the foam pattern width measured at 20 feet distance shall be at least 10 feet for the type I nozzle and 15 feet for the type II nozzle. The fan-shaped water stream pattern width shall be 10 ±2 feet measured at maximum range. The agent discharge direction shall be parallel to the nozzle longitudinal axis. Features shall be incorporated that will secure the pattern selector such that the pattern may not be accidently changed during discharge. If detents are used to secure the selector, bearing surfaces shall be hardened to prevent grooving. While discharging foam, intentional pattern change shall be possible without having to interrupt discharge.

3.5.7 Strength of materials. While lying on a concrete surface, the nozzle shall be capable of supporting the full weight of a man weighing no less than 200 pounds, standing on any point thereon, without causing breakage or permanent deformation.

3.5.8 Hydrostatic pressure. The nozzle shall withstand static pressures up to 750 psig without permanent deformation.

3.5.9 Environmental conditions. The nozzle shall operate satisfactorily under the following conditions:

a. Temperatures ranging from -65° to +125°F

b. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost

c. Exposure to salt-fog

d. Exposure to sand and dust particles as encountered in desert areas.

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3.5.9.1 High temperature. The nozzle shall not be adversely affected when subjected to temperatures up to 450°F.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.7 Dimensions. The nozzle shall be as small as possible consistent with good design and optimum performance. The overall dimensions of the nozzle, including hose fittings, shall be in accordance with figure 1.

3.8 Weight. The weight shall not exceed the following:

a. Type I - 10 pounds

b. Type II - 12 pounds.

3.9 Finishes and protective coatings. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special marking shall be included:

CAPACITY (FOAM SOLUTION):
* GPM

AT 100 PSI NOZZLE PRESSURE

* 60 or 100 GPM, whichever is applicable.

3.11 Workmanship. The nozzle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, welding, brazing, marking of parts and assemblies, painting, riveting, screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect interchangeability, operation, or performance of the nozzle, they shall be held or limited accordingly.

3.11.2 Cleaning. The nozzle shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing of the nozzle shall be classified as follows:

a. Preproduction testing . . . . See 4.4.

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Atmosphere. Unless otherwise specified, nozzle performance, when

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affected by such conditions, shall be demonstrated at or corrected for standard sea level atmospheric conditions of 30 inches Hg barometric pressure, 70°F ambient air temperature, and 50 percent relative humidity.

4.3.2 Item configuration. When subjected to the tests specified herein, the nozzle shall be a complete assembly.

4.3.3 Preparation for test. The nozzle submitted for test shall be properly adjusted and serviced for immediate operation.

4.3.4 Apparatus. Insofar as practicable, apparatus used in conjunction with testing specified herein shall be laboratory precision type and shall be calibrated at proper intervals to assure accuracy.

4.3.5 Operating pressure. The nozzle shall be tested at its design operating pressure, except for the test specified in 4.6.10 and 4.6.11.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. The preproduction test sample shall consist of a model representative of the production items.

4.4.2 Test report. Upon completion of the preproduction tests, three copies of a test report, prepared in accordance with MIL-STD-831, shall be submitted to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events shall be recorded

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any correction to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Materials and information. When nozzles are submitted for test, they shall be accompained by the applicable bill of material and manufacturer's drawings; complete operating instructions; and adequate spare gaskets, locking seals, et cetera, to provide replacements which may be required during the disassemblies and reassemblies specified herein.

4.4.4 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests

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b. Sampling plan and tests.

4.5.1 Individual tests. Each nozzle shall be subjected to the test specified in 4.6.1.

4.5.2 Sampling plan and tests. One nozzle shall be selected at random from every one hundred or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Discharge capacity .... See 4.6.6.1

b. Discharge range ..... See 4.6.6.2

c. Foam quality ..... See 4.6.7

d. Hydrostatic pressure ... See 4.6.10.

4.5.2.1 Rejection and retest. When one nozzle selected from a production run fails to meet the specification, nozzles still on hand or later produced shall not be accepted until the extent and cause of failure are determined.

4.5.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of nozzles on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in nozzles already accepted. Investigation of a test failure could indicate that defects may exist in nozzles already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The nozzle shall be inspected to determine compliance with this specification with respect to material, workmanship, marking, and finishes.

4.6.2 Features and components. The nozzle shall be inspected to determine that all components and features necessary and required are included and properly installed.

4.6.3 Dimensions and weights. The nozzle weight and dimensions shall be checked. This shall be performed on major components wherein no disassembly is required, or as directed by the procuring activity.

4.6.4 Mechanical inspection. A mechanical inspection of all components and parts of the nozzle shall be made and the data recorded as to condition, defects of manufacturer, damage in transit, and damage through use prior to test. The inspection shall include complete servicing of the nozzle to insure readiness for operation.

4.6.5 Functional check. All components, fixtures, controls, and mechanisms shall be moved through their maximum normal displacements and shall show no malfunctioning, objectional rough operation, vibration, or irregular movements.

4.6.6 Performance

4.6.6.1 Discharge capacity. The nozzle shall be discharged at the intended service pressure for a set interval of time for each discharge pattern. The discharge volume shall be measured, and the rate determined for each type discharge. The capacities shall be within ±5 gpm of those specified in 3.5.1.

4.6.6.2 Discharge range. With the nozzle elevated for maximum range in still air, discharge shall be made for a set interval of time for each discharge pattern. The maximum distances from the nozzle to the far side of the ground pattern shall be measured, and shall be not less than those specified in 3.5.2.

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4.6.6.3 Water [illegible]

4.6.6.3.1 Water spray discharge. The nozzle shall be [illegible]eld at a height of 10 feet above the ground and discharged downward for a fixed time interval into sampling pans positioned throughout the ground pattern area. The depth of precipitation shall be measured and the water rate determined.

4.6.6.3.2 Foam pattern. The foam ground pattern shall be determined as described for Hand Line and Auxiliary Nozzles in Standard NFPA No. 412. Foam depths throughout the pattern area shall be converted to liquid application rate. The average water rate shall exceed 0.20 gpm per square foot of ground pattern area.

4.6.7 Foam quality tests. The water used in producing foam for these tests shall have a pH value between 6 and 7.5 at 70°F as determined by means of a standard pH electrometer using the glass electrode.

4.6.7.1 Drainage rate. The foam drainage rate shall be determined as described in Standard NFPA No. 412 using foam samples obtained during the test described in 4.6.6.3.1, and shall be not less than 6 minutes.

4.6.7.2 Expansion. The foam expansion shall be determined as described in Standard NFPA No. 412 using foam samples obtained during the test described in 4.6.6.3.1, and shall be not less than 9.

4.6.8 Fuel agitation. An area 10 feet square with 3-inch-high diked sidewalls, and completely open at the top, shall be constructed to retain liquids. The area shall be filled with water and 50 gallons of gasoline to within 1 inch of the top. The fuel shall be ignited and allowed to preburn for 15 seconds. Foam shall be applied from the nozzle to the burning fuel in the patterns and distances specified below, and burning fuel shall not splash over the sides of the diked area or be so agitated as to cause a flare-up of burning fuel (see 3.5.6):

a. Solid-stream foam - 40 feet

b. Maximum dispersed foam - 20 feet

c. Dispersed foam - 6 feet.

4.6.9 Strength of materials. The nozzle shall be laid on a concrete surface and a man, weighing no less than 200 pounds, shall step on various critical points on the nozzle, i.e., the barrel, valve operating control handle, pattern deflector housing, et cetera, with his full weight supported thereon. There shall be no evidence of breakage or permanent distortion of the nozzle.

4.6.10 Hydrostatic pressure. The valve assembly shall be subjected to a hydrostatic pressure of 750 psig for a 5-minute period, after which it shall be closely examined, and measured if necessary. There shall be no evidence of permanent deformation.

4.6.11 Valve operating test. The valve assembly shall be subjected to a line pressure of 600 psig, and the maximum force required to operate the valve shall be determined and shall not exceed 16 pounds.

4.6.12 Environmental tests. The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810 with exceptions as specified herein. Upon completion of each test, the nozzle shall function properly, and no harmful corrosion or material failure shall be evident.

a. High temperature - Method 501, procedure I, except the nozzle shall be subjected to a temperature of +450°F for a minimum of 2-1/2 minutes. At the end

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of this period, the nozzle shall be removed from the test chamber, inspected for damage, and all movable controls checked for ease of operation. The nozzle shall then be stabilized to and operated at a temperature of 125°F

b. Low temperature - Method 502, procedure I

c. Humidity - Method 507, procedure I

d. Salt-fog - Method 509, procedure I

e. Sand and dust - Method 510, procedure I.

4.6.13 Reliability demonstration. Satisfactory completion of all tests specified herein, without any failure chargeable to the equipment as a reliability failure, demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.14 Final inspection teardown. Upon completion of all tests, the nozzle shall be disassembled to the maximum practical extent for a thorough examination of all parts for failure or undue wear. Any evidence of distortion or deviation from the dimensional requirements herein shall be cause for rejection. The nozzle shall be examined to verify ease of disassembly and assembly, for repair and maintenance, and compliance with 3.4.2.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine compliance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Nozzles shall be preserved and packaged in accordance with MIL-H-775.

5.1.2 Level C. Nozzles shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Levels A and B. Nozzles shall be packed in accordance with MIL-H-775.

5.2.2 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Nozzle, Fire Hose, Foam-Water, Aircraft Fire Fighting, * GPM

*60 or 100 GPM, whichever is applicable.

6. NOTES

6.1 Intended use. The nozzles covered by this specification are intended to be installed on aircraft fire-fighting trucks for use in the rescue of entrapped personnel from crashed and burning aircraft and in combatting other fires involving aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type nozzle required (see 1.2 and 3.10)

c. Location and conditions for preproduction testing (see 4.4)

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d. Level of packaging and packing desired (see section 5).

6.3 Definitions

6.3.1 Foam liquid. Foam liquid is protein base liquid conforming to O-F-555.

6.3.1.1 Foam solution. When the foam liquid is mixed with water for use as a fire extinguishing agent, the solution thus formed is termed a foam solution. The solution used for tests herein will consist of 6 parts of foam liquid to 94 parts of water and will be designated a 6-percent foam solution.

6.3.2 Straight-stream water. Straight-stream water is the term applied to any discharge of water which starts its trajectory in a solid stream.

6.3.3 Fan-shaped water stream. Fan-shaped water stream denotes a thin, triangularly-shaped discharge of homogeneous cross-section and of high velocity.

6.3.4 Water-fog. Water-fog denotes a conically-shaped homogeneous discharge of very minute water droplets, but somewhat larger than the droplets in natural fog.

6.3.4.1 Water-fog pattern. The contiguous cross-sectional area formed by projecting the homogeneous, translucent, or semiopaque conical discharge upon a backboard parallel to the axis of the discharge is termed the water-fog pattern area for test purposes.

6.3.5 Solid-stream foam. When the foam discharge flows through the air at the start of the trajectory in such a manner as to appear to be a solid cylinder, the flow is considered to be solid-stream foam.

6.3.6 Dispersed foam. When foam is discharged in such manner as to form flakes similar in appearance to snowflakes, such discharge is designated as dispersed foam. This foam is of the same general quality as solid-stream foam.

6.3.7 Maximum discharge range. The maximum range for foam discharge will be the distance from the nozzle to the far side of the discharge ground pattern. The maximum range of a water-fog discharge will be the maximum horizontal distance at which the discharge is translucent and has an appearance similar to that of a jet of steam.

6.3.8 Water application rate. The discharge of water or foam solution into the pattern area (see 6.3.11) will be expressed in gallons per minute per square foot of pattern area which will denote the water application rate.

6.3.9 Drainage time. The time required for 25 percent of the liquid to drain out of the foam mass is denoted as the drainage time.

6.3.10 Expansion. The expansion value is the ratio of the volume of the foam to the volume of the original solution before adding air. It is the numerical value of the reciprocal of the specific gravity of the foam.

6.3.11 Ground pattern. The ground pattern is the contiguous flat surface area upon which the discharge falls, when the nozzle is held to obtain maximum range, and upon which the discharge rate of water exceeds 0.05 gpm per square foot of area.

6.3.12 Foam blanket. Foam blanket denotes the covering over the area including the discharge ground pattern and the adjacent area to which the foam flows on a level surface.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Reviewers:
Air Force - 11, 84, 85

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8979

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